

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4				Indicative content: Dark lines: Certain wavelengths of light are absorbed by gas atoms in the atmosphere of the star and are re-emitted in all directions so appear dimmer. Each element has its own specific dark line(s). Elements: Helium is present because there are dark lines at 447 and 502 nm. Hydrogen is present because there are dark lines at 410, 434, 486 and 656 nm. Sodium is present because there is a dark line at 590 nm. Iron is not present because there are no dark lines at 431, 467, 496 and 527 nm. [Another element may be present because of the line at 422 nm.] N.B. References to red shift and Big Bang are neutral. 5–6 marks Detailed description of both areas. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Detailed description of one area or limited descriptions of both. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	4	2		6	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1–2 marks A limited description of one area or very limited description of both. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> <i>The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
Question 4 total					4	2	0	6	2	0